asclib를 react native에서 메소드 바인딩 하기

테스트 환경

node버전: 20

react 버전: 19.2.3

react-native 버전: 0.84.1

acslib 메소드 바인딩하기

1. `ReactContextBaseJavaModule` 생성

<root>/android/app/src/main/java/com/<packageName>/AcsLibModule.java 코드

```java
public class AcsLibModule extends ReactContextBaseJavaModule {

    private AcsCoordinateConverter acsCoordConverter = ACS.coordinateConverter().create();
    private static final String MODULE_NAME = "AcsLibModule";

    private Context mContext;

    public AcsLibModule(ReactApplicationContext reactContext) {
        super(reactContext);
    }

    @NonNull
    @Override
    public String getName() {
        return MODULE_NAME;
    }

    @ReactMethod
    public void format(double lat, double lon, String coordTypeStr, Promise promise) {
        try {
            CoordinateType coordType = CoordinateType.from(coordTypeStr);
            String coord = acsCoordConverter.format(lat, lon, coordType);
            promise.resolve(coord);
        } catch (Exception e) {
            promise.reject("CONVERSION_ERROR", "Failed to convert coordinates: " + e.getMessage());
        }
    }
    // 나머지 api 브릿지 구현
}
```

2. `ReactPackage` 생성, 1번에서 만든 Module을 불러오기

<root>/android/app/src/main/java/com/<packageName>/AcsLibPackage.java 코드

```java
import com.facebook.react.ReactPackage;
import com.facebook.react.bridge.NativeModule;
import com.facebook.react.bridge.ReactApplicationContext;
import com.facebook.react.uimanager.ViewManager;

import java.util.Arrays;
import java.util.Collections;
import java.util.List;

public class AcsLibPackage implements ReactPackage {

    @Override
    public List<NativeModule> createNativeModules(ReactApplicationContext reactContext) {
        return Arrays.asList(new AcsLibModule(reactContext));
    }

    @Override
    public List<ViewManager> createViewManagers(ReactApplicationContext reactContext) {
        return Collections.emptyList();
    }
}
```

3. MainApplication 에 2번에서 만든 `ReactPackage` 추가

<root>/android/app/src/main/java/com/<packageName>/MainApplication 코드 수정

```
...
packageList =
  PackageList(this).packages.apply {
    add(AcsLibPackage()) // AcsLibPackage를 add하기
  },
...
```

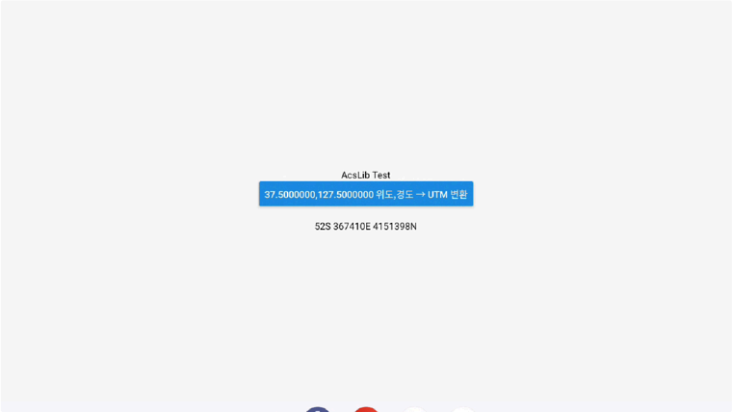
4. AcsLibModule불러오기, react component에서 호출

AcsLib.ts

```ts
const { AcsLibModule } = NativeModules;
if (!AcsLibModule) {
  throw new Error(
    'AcsLibModule not found. ' +
    'Make sure AcsLibPackage is registered in MainApplication.java ' +
    'and you have rebuilt the app.'
  );
}

const AcsLib = {
  format(
    lat: number,
    lon: number,
    coordType: string
  ): Promise<string> {
    return AcsLibModule.format(lat, lon, coordType);
  },
  // AcsLibModule에서 ReactMethod로 만든 method를 더 추가
};

export default AcsLib;
```

App.tsx 예시

```tsx
import React, { useState } from 'react';
import { View, Text, Button } from 'react-native';
import AcsLib from './AcsLib';

export default function App() {
  const [result, setResult] = useState<string>("");

  const handleConvert = async () => {
    try {
      const coord = await AcsLib.format(
        37.5000000, // 위도
        127.5000000, // 경도
        "UTM"
      );
      setResult(coord);
    } catch (e) {
      console.error(e);
      setResult("변환 실패");
    }
  };

  return (
    <View style={{ flex: 1, justifyContent: 'center', alignItems: 'center' }}>
      <Text>AcsLib Test</Text>

      <Button
        title="37.5000000,127.5000000 위도,경도 → UTM 변환"
        onPress={handleConvert}
      />

      <Text style={{ marginTop: 20 }}>
        {result}
      </Text>
    </View>
  );
}
```

demo

아래는 acslib의 위경도 변환 메소드(format)를 호출한 결과이다.